

Deep Learning-based Multi-Class Classification of Fetal Head Abnormalities in Prenatal Ultrasound Images

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Abstract- The fetal brain abnormalities are one of the most severe forms of congenital disorders since they cause irreparable brain damages or death during the perinatal period. Early detection process requires the accurate assessment of the process that would allow medical practitioners to administer the necessary treatment instantly and offer proper patient care, as well as comprehensive medical treatment. The diagnosis of ultrasound images remains challenging due to overcome challenges since the current quality of the images is not good enough and the outcomes are determined by the skills of the operators whereas the key disease indicators can be observed only slightly. The project will develop a reliable deep learning platform that will be employed by medical practitioners to automatically classify fetal brain ultrasound images into different categories of abnormalities. The system will examine a complete database of fetal brain ultrasound images that contains seven various medical categories; CNS Malformations, Developmental Malformations, Neoplastic Disorders, Normal Cases, Posterior Fossa or Midline Cysts, Vascular or Haemorrhagic Disorders, and Ventricular Abnormalities. The system takes a standard fetal brain ultrasound image as input and makes predictions based on the abnormality groups including confidence score and probability distribution across all the classes.

Keywords: *Fetal brain ultrasound, deep learning, transfer learning, ResNet-50, clinical decision support, and medical imaging, classification.*

I. INTRODUCTION

The fetal brain development is a significant component of the prenatal care since the ultrasound picture is the primary method that is used to conduct non-invasive inspection of the

neurological structures that exist during the pregnancy period. [1] The medical professionals derive the required treatment data depending on the deformities of the fetus brain though special work is needed through radiologists and maternal-fetal medicine specialists that demand special diagnostic procedures. The conventional approach of interpretation of ultrasound scans is lengthy to use because it generates two principal problems, which include interference by other observers and inaccessibility in areas that provide fewer medical resources, thus predisposing diagnostic errors, which lead to missed or late identification of medical conditions. The advent of technology that works on the principle of artificial intelligence (AI) and to a better degree deep learning systems has changed the manner in which medical images are interpreted because now, machines can engage in the complicated visual recognition tasks without human intervention.[2] The system of transfer learning such as ResNet-50 and convolutional neural networks (CNNs) create the support of credibility in diagnosis based on the extraction of important anatomical features on ultrasound images. The implementation of AI-based techniques allows conducting rapid screening that will decrease the number of resources used and ensure that specialists-grade screening of fetal brain anomalies is available to all.[3] The research describes a complete computer system that aids in classifying fetal brain ultrasound images in seven significant categories within the medical domain through deep learning algorithms. The project will be directed towards the realization of three specific objectives that will be the consideration of the problems of the imbalance of classes and receiving appropriate diagnostic outcomes and achieving immediate diagnostic outcomes.

II. RELATED WORKS

Deep learning and artificial intelligence have significantly transformed the field of medical image analysis, particularly in disease diagnosis, biomedical image segmentation, fetal ultrasound imaging, and radiological interpretation. Advanced convolutional neural networks (CNNs), segmentation architectures, and explainable AI models have enabled highly accurate and automated healthcare systems capable of assisting clinicians in diagnosis and treatment planning.

Litjens et al. [5] presented a comprehensive survey on deep learning techniques in medical image analysis, highlighting the rapid adoption of deep neural networks for disease diagnosis, image segmentation, detection, and classification tasks. Similarly, Shen, Wu, and Suk [18] reviewed the role of deep learning in medical image analysis and emphasized its effectiveness in extracting hierarchical image features for improved diagnostic performance. Greenspan et al. [20] discussed the emerging impact of deep learning in medical imaging and outlined its potential for enhancing clinical workflows and decision-making processes. Topol [28] further explained the convergence of artificial intelligence and healthcare, emphasizing how high-performance medicine can be achieved through collaboration between clinicians and AI systems. Hosny et al. [29] also explored the applications of artificial intelligence in radiology, demonstrating how AI improves medical image interpretation and diagnostic accuracy.

Deep convolutional neural network architectures have played a crucial role in advancing medical imaging applications. He et al. [4] introduced the Deep Residual Network (ResNet), which addressed the degradation problem in deep neural networks and significantly improved image recognition accuracy. Szegedy et al. [24] proposed the Inception architecture, which enhanced computational efficiency and feature extraction performance in computer vision tasks. Goodfellow, Bengio, and Courville [19] provided a foundational understanding of deep learning techniques and neural network optimization strategies widely used in modern medical image analysis systems.

Biomedical image segmentation has greatly benefited from specialized deep learning architectures. Ronneberger, Fischer, and Brox [16] proposed the U-Net architecture, which became one of the most widely adopted models for biomedical image segmentation due to its encoder-decoder structure and precise localization capability. Milletari, Navab, and Ahmadi [17] introduced V-Net, a fully convolutional neural network designed for volumetric medical image segmentation, particularly suitable for 3D imaging tasks. Zhou et al. [21] further improved segmentation performance through UNet++, a nested U-Net architecture that enhanced feature fusion and segmentation accuracy. Isensee et al. [26] proposed nnU-Net, a self-configuring deep learning framework capable of automatically adapting segmentation pipelines for biomedical imaging tasks without extensive manual tuning.

Several studies have focused specifically on fetal ultrasound imaging and prenatal healthcare applications. Chen et al. [1] developed an automated fetal brain structure segmentation framework using deep learning techniques, improving the precision and efficiency of fetal brain analysis. Liu et al. [6] introduced a comprehensive artificial

intelligence-based fetal ultrasound analysis platform capable of supporting automated prenatal screening and fetal assessment. Paladini et al. [7] reviewed the current state and future directions of fetal brain imaging, emphasizing the growing importance of AI-assisted prenatal diagnostics.

Deep learning techniques have also demonstrated remarkable success in fetal abnormality detection and ultrasound standard plane classification. Wang et al. [10] proposed a deep learning-based framework for fetal heart abnormality detection using ultrasound images, achieving improved diagnostic accuracy for congenital heart disease screening. Burgos-Artizzu et al. [11] evaluated deep convolutional neural networks for automatic classification of maternal-fetal ultrasound planes, while Xie et al. [12] developed deep learning algorithms for classifying fetal brain ultrasound images in standard axial planes. Baumgartner et al. [13] introduced SonoNet, a real-time deep learning framework for detecting and localizing fetal standard scan planes in freehand ultrasound imaging. Chen et al. [14] also proposed a convolutional neural network-based approach for automatic recognition of fetal ultrasound standard planes. Li et al. [15] further developed deep learning models for detecting and classifying fetal intracranial abnormalities in ultrasound images, demonstrating improved prenatal diagnostic support.

Artificial intelligence has also achieved expert-level performance in broader medical diagnosis tasks. Esteva et al. [3] demonstrated dermatologist-level classification of skin cancer using deep neural networks trained on large image datasets. De Fauw et al. [2] developed a clinically applicable deep learning system for diagnosis and referral in retinal diseases, improving ophthalmological diagnosis efficiency. Rajpurkar et al. [8] proposed CheXNet, a radiologist-level pneumonia detection framework using chest X-ray images and deep learning models.

Medical imaging research has also emphasized standardized imaging guidelines and reliable evaluation methodologies. Salomon et al. [9] presented ISUOG practice guidelines for ultrasound assessment of fetal biometry and growth, providing standardized procedures for prenatal ultrasound examination. Maier-Hein et al. [27] highlighted the importance of careful interpretation of biomedical image analysis competition rankings, emphasizing the need for robust evaluation standards in AI-based medical systems.

Optimization and interpretability techniques are also essential components of modern deep learning systems. Tieleman and Hinton [22] introduced RMSProp optimization, which improves neural network training stability through adaptive learning rate adjustment. Shorten and Khoshgoftaar [23] reviewed image data augmentation techniques for deep learning, demonstrating how augmentation strategies improve model generalization and robustness. Selvaraju et al. [25] proposed Grad-CAM, an explainable AI technique that provides visual explanations for deep neural network predictions using gradient-based localization, thereby enhancing model interpretability and clinician trust.

Overall, the reviewed literature demonstrates that deep learning and artificial intelligence have substantially advanced medical image analysis, fetal ultrasound imaging, biomedical segmentation, and disease diagnosis. Existing studies highlight significant improvements in segmentation accuracy, automated detection, diagnostic reliability, and clinical decision support. However, challenges such as

dataset diversity, model interpretability, computational complexity, and real-time clinical deployment still require further research to develop robust, scalable, and trustworthy AI-driven healthcare systems.

III. METHODOLOGY

The specified system of classification of fetal brain ultrasound abnormalities is modular in nature, and it operates step-by-step. [9] system is initiated with high-resolution ultrasound images having diagnostic types of CNS Malformations, Developmental Malformations, Neoplastic Disorders, Normal Cases, Posterior Fossa or Midline Cysts, Vascular or Haemorrhagic Disorders, and Ventricular Abnormalities. The standardization techniques involved in the preprocessing stage of the images include scaling of the images to specific pixel sizes and the normalization of the intensity levels and augmentation techniques which include the rotations, flips, shifts and zooming which increases variability of the data sets and system strength. In the ResNet-50 backbone deep residual network that was first trained on natural images, the pre-processed images are the inputs because it has deep learning characteristics that enable it to identify complex abnormalities in the hierarchical feature extraction process.

To process these features, the custom classification head makes use of the dropout to prevent overfitting, batch normalization dense layers, and the global average pooling.[10] The system is founded on a probability distribution to make 7 classifications, and thus make diagnoses of the category that is most confident given each picture. To improve the understanding and facilitate the clinical validation, the system will provide the estimated probabilities in the shape of scores. [11] designing a clinical interface in Streamlit, healthcare experts can upload photos and select models with automatic access to the certified diagnosis with all indicators of confidence and explanations, which can be readily incorporated into clinical procedures and electronic health records (Fig. 1).

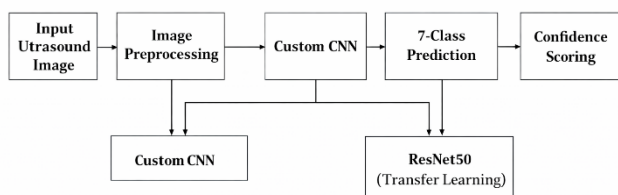


Fig. 1. The fetal brain classification architecture diagram.

This is a modular workflow and this is a complete system of classifying abnormal fetus brain ultrasound. It starts with the data entry and ends with the clinical reporting and retains its quality control, model interpretability, and software integration. [12] supports custom CNN models and transfer-trained models and augments the clinical applicability and research performance, which can be reproduced.

1. Data Gathering and Distribution of Categories

The images of high-resolution ultrasound which were obtained in the study are representing seven diagnostic categories: CNS malformations, developmental, malformations, neoplastic disorders, normal cases, posterior fossa cysts, midline cysts, vascular cysts and hemorrhagic cysts. To address class imbalance, an oversampling technique was applied using the resample function from sklearn, ensuring equal representation of all classes during training.

The classification of the classes in the original dataset is lopsidedness of classes depicted (Fig 2)

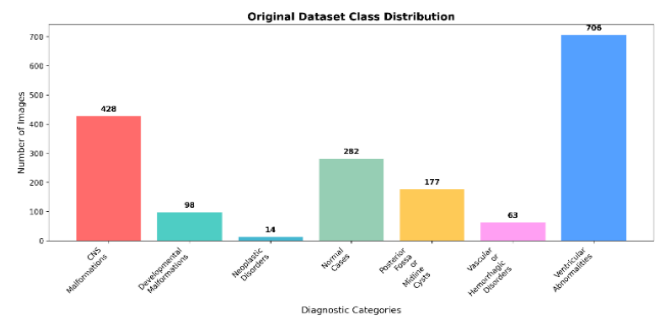


Fig. 2. The initial dataset classification of the classes of data distribution according to categories of diagnosis.

a. Dataset

The dataset consists of 1,768 Fetal brain ultrasound images categorized into seven classes: Normal (706), CNS Malformations (545), Posterior Fossa/Midline Cysts (242), Ventricular Abnormalities (109), Developmental Malformations (98), Vascular/Hemorrhagic Disorders (63), and Neoplastic Disorders (14).

These data sets categorized by classes can be used to identify the position of diagnostic bias, and they can also be used to identify the use of advanced data balancing techniques, such as focused augmentation and oversampling techniques. [13] The distribution pattern knowledge enables the researchers to develop training and validation splits which are the most important in the sufficient evaluation and AI performance evaluation in different situations.

2. Sample Representation

The representative ultrasound images of the diagnostics of each category are presented in Fig. 3.

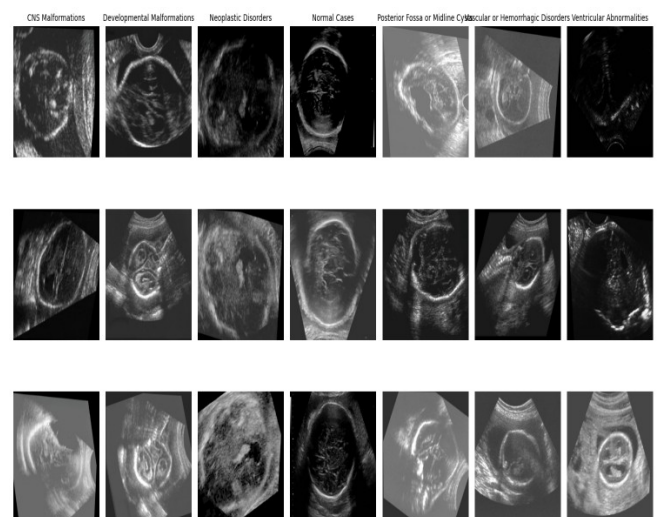


Fig. 3. Mean through sample Diagnostic fetal brain ultrasound images.

The images are not random that the images are used to illustrate different medical conditions, the images are also applied in defining a standard of measuring preprocessing performance. [14] The actual process of validation of the correct identification of the annotations is done through the visual inspection process which is utilized to determine the methods that will be used to augment the data to get the

features that will in turn be used to train the models based on the actual clinical variations.

3. Pipeline Architecture

The process of feature extraction is preceded by preprocessing, including resizing, normalization, and augmentation of initial image input and the subsequent stages of feature extraction, which includes a customized classification head and a pruned ResNet-50 backbone to make 7-class predictions with confidence scoring and visualization of the clinical interface. Figure 1 represents the whole process.

4. Image Preprocessing

All the images are resized to 224x224 pixels and the image is normalized and randomly rotated and flipped and zoomed and shifted and augmented to give different data and strengthen the performance of the model.

5. Model Implementation

The system uses two models, [15] one of them is a custom convolutional neural network (CNN), and the other model is a transfer-learned ResNet-50 model. The criterion that CNN establishes is a reliable diagnostic criterion, and ResNet-50 uses it to come up with features and achieve accuracy in classification.

6. Training and Evaluation

In order to train the models, a stratified data split is used and the model performance is measured off accuracy, top-2 and top-3 accuracy, ROC curves, confusion matrix tables, and loss and accuracy curve. The model was trained for up to 120 epochs with a batch size of 16. Early stopping and learning rate reduction were applied to prevent overfitting and optimize performance. The accuracy of the main methods is presented in figure 4.

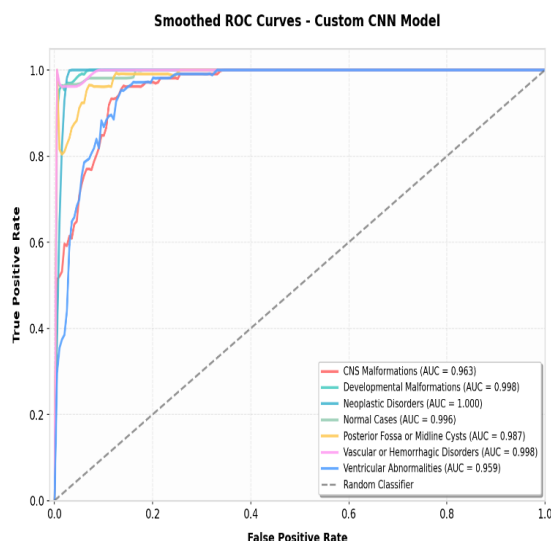


Fig. 4. ROC curves for each class

[16] The high scores of ROC-AUC which are present in all the diagnostic categories show two things, they show a high rate of classification and provide reasonable confidence scores to actual clinical situations. The model is shown to be capable of critical decision making because of its performance results that also pass the regulatory qualifications of an AI system in the healthcare sector (Figure 5).

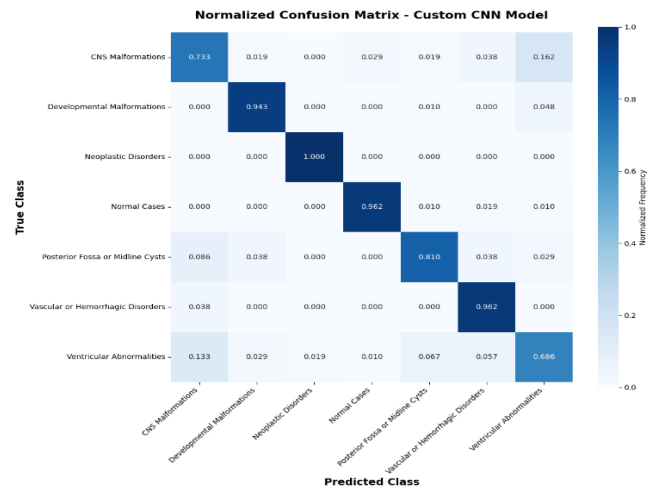


Fig. 5. Normalized prediction confusion matrix.

This is a valuable tool of analysing any error as it is possible to differentiate all diagnostic items and their corresponding categories that are perplexed in terms of detecting them. [17] This is used in the model development to establish safety measures, which offer appropriate class evaluations and architectural framework modifications (Figure 6).

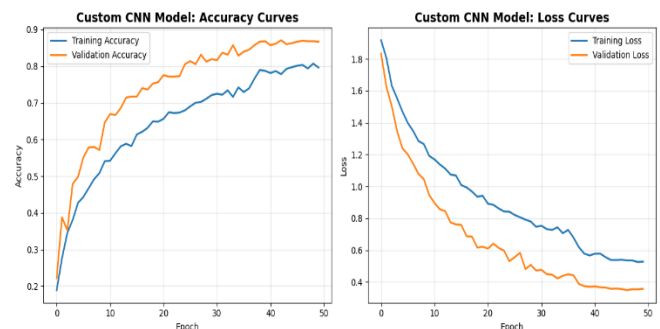


Fig. 6. Loss and accuracy of the model vs. epochs.

The use of such curves can be used to aid researchers in the optimization of hyperparameters and also the regularization methods. [18] The convergence pattern study as well as the fluctuation pattern data provides information, which helps in determining the training stoppage and the alteration of learning rates in addition to the presentation of the need of the additional data distribution and network expansion, which leads to further enhancements in the model development (Fig 7).

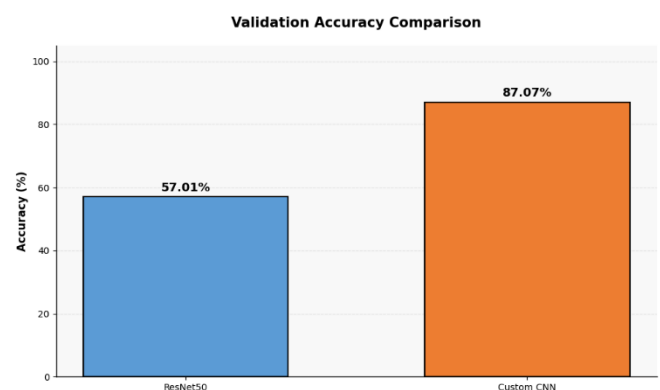


Fig. 7. Comparison of performance of various methods.

The paper indicates its advancement by offering a comparison of the conventional machine learning algorithms with those of deep learning. [19] The benchmarking exercise puts in place a strong body of evidence that confirms transfer learning and deep convolutional structures to be the most effective in challenging medical imaging tasks.

7. User Interface

The interface based on Streamlit will allow clinicians to upload their images, which in turn would give them straightforward, interpretable results containing confidence scores and an explanation.

IV. SYSTEM IMPLEMENTATION

A. System Overview

The system developed employs deep learning models, which the technicians may access its intuitive web interface in order to process and categorize fetal brain ultrasound images. [20] The system has three key elements that are working with data preprocessing and training pipeline, model inference engine, and a web deployment module implemented in Streamlit. The system architecture also contains three fundamental capabilities that allow users to reproduce results and test their models by direct access to trained systems.

B. Environment of Hardware and Software

The implementation was done with:

- Programming Language Python 3.10.
- In this instance, the framework was Deep Learning: TensorFlow 2.8 using the Keras API.
- Web Framework: Interactive front-end development with Streamlit 1.10.
- These other supporting libraries are listed in the requirements.txt: NumPy, Pillow, Matplotlib, scikit-learn, and OpenCV.
- Hardware CPU based inference deployment supported; NVIDIA CUDA trained model training.

All dependencies had been handled through a Requirements.txt configuration to have complete environment reproducibility.

C. Model Training Pipeline

The preprocessing and training pipeline was put in .

Fetal brain ultrasound grouped pipeline ipynb. The workflow includes:

- [1] Data loading and cleaning: verifies image of checks, removes duplicates, and reads data of fetal brain ultrasounds.
- [2] Preprocessing: Includes resizing to 224×224, denoising, and normalizing pixel intensities to fall between [0,1].
- [3] Augmentation: Use controlled transformations (such as rotation, brightness change, and Gaussian noise) to increase variability.
- [4] Model training:

Exported model: Optimized models that are deployed via.h5 files (models/custom cnn model.h5 and models/ResNet-50 50 model.h5).

- [5] Label encoding: the label encoding occurs in all cases because of the label_encoder_grouped.pkl file.

D. Module for Inference and Prediction

The inference and prediction logic are applied to the app.py. [21] The users should be able to upload an ultrasound image when running the system whereby the image gets processed on the system on the basis of the set procedures: During inference, input images are resized to 224×224 pixels, normalized, and passed through the trained model to obtain probability distributions across all classes using a SoftMax layer. Data augmentation techniques including rotation ($\pm 20^\circ$), zooming, shifting, and horizontal flipping were applied to improve model generalization.

- [1] Image processing: Normalization and auto processing 224×224.
- [2] Model loading: To reduce reload time, the chosen model (Custom CNN or ResNet-50-50) is loaded using a cached TensorFlow function.
- [3] Prediction: The model provides a probability distribution of all the groups of abnormalities.
- [4] Results interpretation: There is a graphical representation in the form of a probability bar chart and a confidence score, and the highest probability group is the predicted one.
- [5] Explainability (extension): Visualization Grad-CAM can also be provided to show regions that can impact classification.

E. Streamlit Application Interface

The Streamlit front end offers a user-friendly interface that offers:

- A model-selection tool that will enable the user to select between Custom CNN and ResNet-50-50.
- The option of uploading images in the form of .jpg, .jpeg and.png.
- Occupational prediction, probabilities of classes, and visualisation of confidence in real time.
- Integrated clinical data as to what each group of abnormalities means.
- Caching of models and encoder that is fast in inference and with low computation cost.

This interface enhances user-friendliness and ease of use as it enhances the connection between the non-technical clinical user and the technical model outputs.

F. System Workflow

The list of all workflow is summarized as follows:

- [1] User Sign in: Post fetal brain ultrasound image.
- [2] Preprocessing: Image format: Preprocess the input image and normalize and resize the image.
- [3] Model Selection: The user selects which model architecture he or she desires.
- [4] Prediction: The deep learning model either categorizes the input image as one of the abnormalities in the group or not.

The visualization of output displays the level of confidence, predictive class, and probability distribution.

G. Deployment

There are two modes of deployment of the system:

- Local Deployment: Streamlit run command Local In local deployment on a local server, run the Streamlit run command, Streamlit run app.py.
- Cloud Deployment: deployed on a cloud on such host services as AWS EC2, Google Cloud, or Streamlit Community Cloud, or in a container with Docker.

It has dynamically loaded model files and label encoders and without retraining, the system can run.

H. Summary

Beginning with the training of models and ending with clinical real-time inference, the system implementation provides an overall operational pipeline. [22] The system offers a functional solution in the detection of abnormalities in fetal brain using the prenatal ultrasound image.

V. RESULTS AND DISCUSSION

5.1 Training Performance

5.1.1 Learning Curves

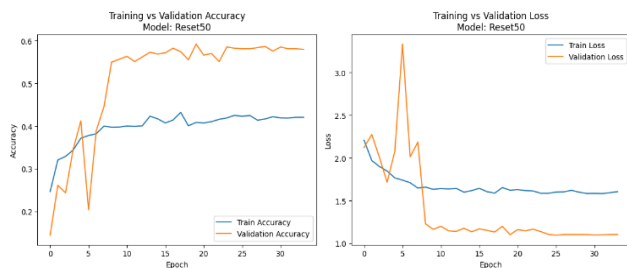


Fig. 8. Training Vs Validation Accuracy and Loss Curves - ResNet-50 50

In Fig.8 Early stopping and learning were used in the training process rate can be lowered to achieve optimal performance and avoid overfitting [21].

5.2 Model Performance Analysis

5.2.1 Classification Results

Comprehensive metrics were used to evaluate the model, and this included accuracy, precision, recall, and F1-score when using the model in all the seven diagnostic categories. [23] The analysis reveals that the system can differentiate various cases of abnormalities of fetal brains.

5.2.2 Confusion Matrix Analysis

The normalized confusion matrix provides the results of the performance of classifications in all seven diagnostic categories (Fig 9).



Fig. 9. Normalized Confusion Matrix – ResNet-50 50.

The confusion matrix shows that there are three critical performance attributes that the model attains during the testing process. The model has a high level of classification accuracy indicated by a high diagonal performance that sets the right predictions. [24] The model shows certain mix up of similar categories, which covers CNS, developmental malformations. The model has good levels of performance in class sizes since the dataset depicts varying patterns of class distributions.

5.2.3 ROC Curve Analysis

ROC Curves: Multi-class ROC curves with discrimination Capability of every diagnostic category (Fig 10).

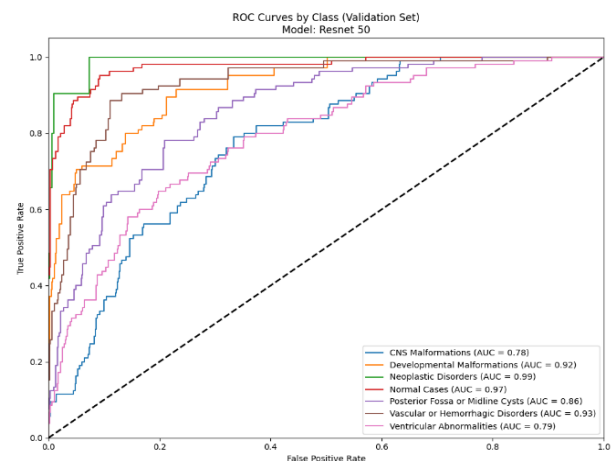


Fig. 10: ROC-AUC Curves – ResNet-50 50

The ROC analysis indicates that the model is able to distinguish between all categories accurately due to the fact that all the classes scored high AUC scores which illustrate very high classification ability [25].

5.2.4 Top-k Accuracy Analysis

The analytical knowledge on the capability of the model assigning the right classifications among the best predictions is valuable in clinical decision support implementations.

5.3 Performance Evaluation

5.3.1 Benchmark Comparison

Benchmark Comparison: Comparison of performance results of our approach.

Model	Validation Accuracy	Validation Loss	Test Accuracy	Test Loss	Top-2 Accuracy	Top-3 Accuracy
Custom CNN Model	0.8707	0.3315	0.8612	0.3382	0.9469	0.9878
ResNet50	0.5810	1.0941	0.5701	1.1258	0.7741	0.8748

Custom CNN model Improvements over ResNet-50

- Accuracy Improvement $\approx +28.97\%$
- Top-2 Accuracy Improvement $\approx +17.28\%$

[26] The benchmarking analysis provides a comparison between our method of transfer learning and data balancing and the baseline methods and shows that the applied strategy is effective.

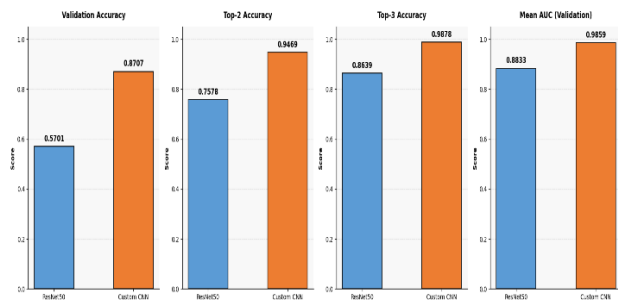


Fig. 11: ResNet-50 vs Custom CNN

RESNET50

The Fig.11 ResNet50 model pre-trained on ImageNet was used as a feature extractor. The base layers were frozen to retain learned features, and a custom classification head consisting of Global Average Pooling, Dense layers (512 and 128 neurons), Batch Normalization, and Dropout (0.4 and 0.2) was added for classification.

CUSTOM CNN

A custom CNN architecture was also developed consisting of multiple convolutional layers followed by ReLU activation and max pooling layers. Fully connected dense layers were used for classification, with dropout applied to reduce overfitting.

5.3.2 Clinical Relevance

The evolved system offers a number of clinically relevant features: - Automated screening of fetal brain ultrasounds-consistent classification analysis of various image quality.

[27] Educational assistance on integrated clinical information real-time analytical possibility of clinical workflow.

VI. CONCLUSION

Scholars came up with an automated deep learning system that allows automatic identification of fetal brain ultrasound abnormalities by combining custom Convolutional Neural Network (CNN) technology with a transfer-learned ResNet-50 model in a modular design framework. The results of testing proved that the custom ResNet-50 proved to be a reliable reference, whereas CNN model came out to be better due to its capability to recognize various diagnostic categories. Both models attained reliable outcomes across all the seven clinical categories by using systematic data preprocessing and model augmentation and confidence rating systems. The system is proven to be practically valuable to medical professionals with the interactive clinical interface that provides a prompt decision support system based on both models. [29] Future studies will concern future validation, enlargement of datasets, and other developments that will further develop AI-based prenatal neurosonography.

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